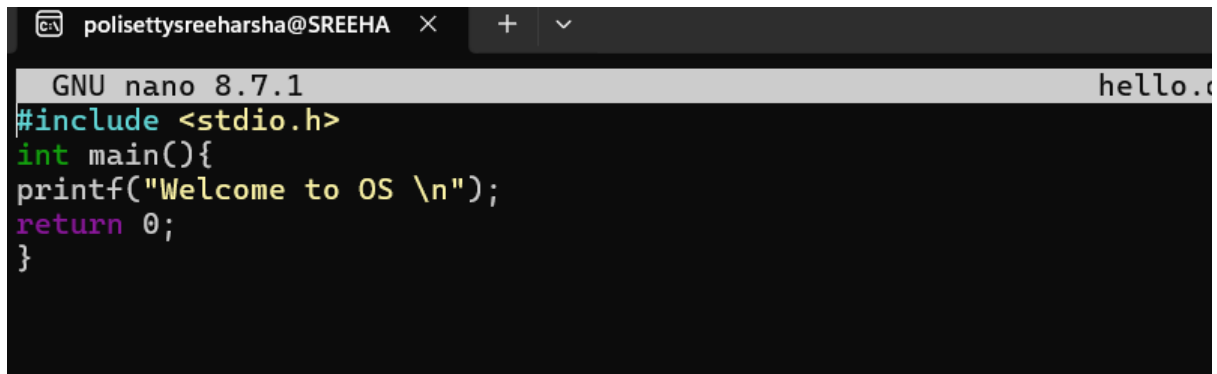


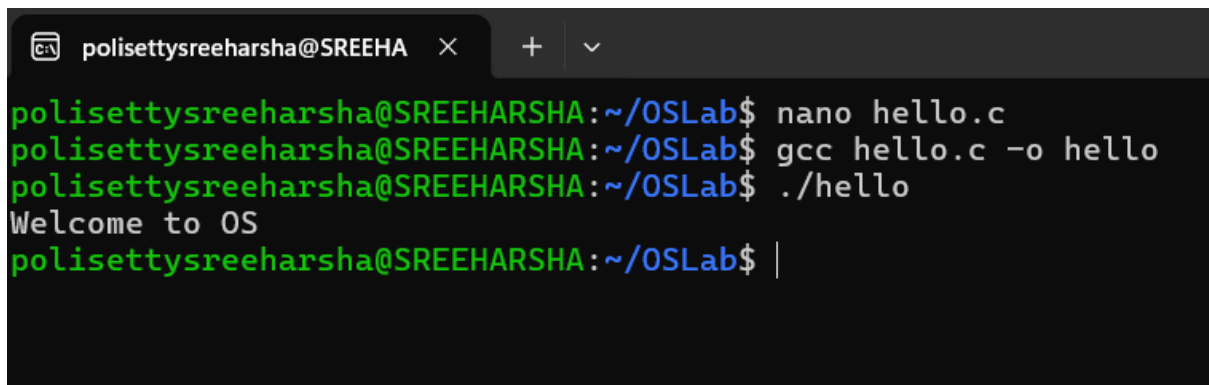
## Experiment 1: HELLO WORLD

Code:

A screenshot of a terminal window with a dark background. The title bar shows 'polisettysreeharsha@SREEHA' and a tab icon. The terminal content shows the GNU nano 8.7.1 editor editing a file named 'hello.c'. The code is as follows:

```
GNU nano 8.7.1 hello.c
#include <stdio.h>
int main(){
printf("Welcome to OS \n");
return 0;
}
```

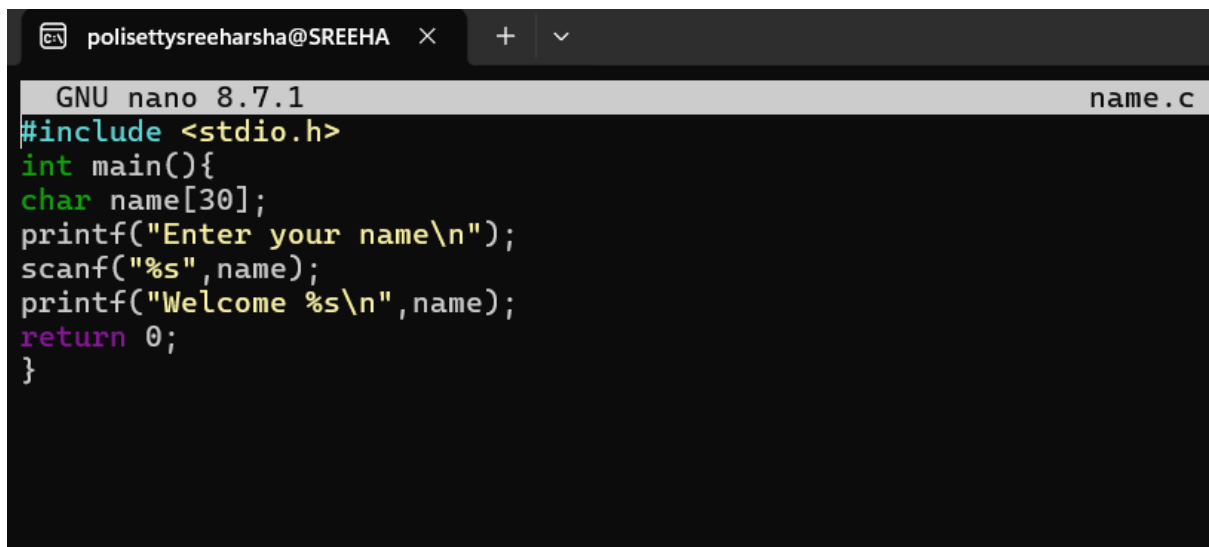
Output:

A screenshot of a terminal window with a dark background. The title bar shows 'polisettysreeharsha@SREEHA' and a tab icon. The terminal content shows the following commands and output:

```
polisettysreeharsha@SREEHARSHA:~/OSLab$ nano hello.c
polisettysreeharsha@SREEHARSHA:~/OSLab$ gcc hello.c -o hello
polisettysreeharsha@SREEHARSHA:~/OSLab$ ./hello
Welcome to OS
polisettysreeharsha@SREEHARSHA:~/OSLab$ |
```

## Experiment 2: INPUT CONSIDERATION

Code:

A screenshot of a terminal window with a dark background. The title bar shows 'polisettysreeharsha@SREEHA' and a tab icon. The terminal content shows the GNU nano 8.7.1 editor editing a file named 'name.c'. The code is as follows:

```
GNU nano 8.7.1 name.c
#include <stdio.h>
int main(){
char name[30];
printf("Enter your name\n");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
polisettysreeharsha@SREEHA x + v
polisettysreeharsha@SREEHARSHA:~/OSLab$ nano name.c
polisettysreeharsha@SREEHARSHA:~/OSLab$ gcc name.c -o name
polisettysreeharsha@SREEHARSHA:~/OSLab$ ./name
Enter your name
Sreeharsha
Welcome Sreeharsha
polisettysreeharsha@SREEHARSHA:~/OSLab$ |
```

Experiment 3: GENERATING ID

Code:

```
GNU nano 8.7.1 Display.c
#include <stdio.h>
#include<unistd.h>
int main(){
printf("Process ID =%d\n",getpid());
return 0;
}
```

Output:

```
polisettysreeharsha@SREEHARSHA:~/OSLab$ nano Display.c
polisettysreeharsha@SREEHARSHA:~/OSLab$ gcc Display.c -o Display
polisettysreeharsha@SREEHARSHA:~/OSLab$ ./Display
Process ID =1112
```

Experiment 4: FORK

Code:

```
GNU nano 8.7.1 Fork.c
#include <stdio.h>
#include <unistd.h>
int main(){
fork();
printf("Hello\n");
return 0;
}
```

Output:

```
polisettyreearsha@SREEHARSHA:~/OSLab$ nano Fork.c
polisettyreearsha@SREEHARSHA:~/OSLab$ gcc Fork.c -o Fork
polisettyreearsha@SREEHARSHA:~/OSLab$ ./Fork
Hello
Hello
polisettyreearsha@SREEHARSHA:~/OSLab$ |
```

## Experiment 5: DETERMINING CHILD OR PARENT PROCESS

Code:

```
polisettyreearsha@SREEHA x + v
GNU nano 8.7.1 parent_child.c
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid;
    pid = fork();
    if(pid==0){
        printf("I am child process\n");
    }
    else{
        printf("I am parent process\n");
    }
    return 0;
}
```

Output:

```
polisettyreearsha@SREEHARSHA:~$ gcc parent_child.c -o parent_child
polisettyreearsha@SREEHARSHA:~$ ./parent_child
I am parent process
I am child process
polisettyreearsha@SREEHARSHA:~$ ./parent_child
I am parent process
I am child process
```

## Experiment 6: PARENT PID , CHILD PID

Code:

```
polisetlysreeharsha@SREEHA x + v
GNU nano 8.7.1 parent_child2.c
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid;
    pid=fork();
    if(pid==0){
        printf("Child pid=%d\n",getpid());
        printf("Parent pid=%d\n",getpid());
    }
    else{
        printf("Parent pid=%d\n",getpid());
    }
    return 0;
}
```

Output:

```
polisetlysreeharsha@SREEHARSHA:~$ nano parent_child2.c
polisetlysreeharsha@SREEHARSHA:~$ gcc parent_child2.c -o parent_child2
polisetlysreeharsha@SREEHARSHA:~$ ./parent_child2
Parent pid=767
Child pid=768
Parent pid=768
```

Experiment 7: WAIT

Code:

```
polisetlysreeharsha@SREEHA x + v
GNU nano 8.7.1 wait.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    pid_t pid;
    pid = fork();
    if(pid==0){
        printf("Child running\n");
    }
    else{
        wait(NULL);
        printf("Parent running after child\n");
    }
    return 0;
}
```

Output:

```
polisettysreeharsha@SREEHARSHA:~$ nano wait.c
polisettysreeharsha@SREEHARSHA:~$ gcc wait.c -o wait
polisettysreeharsha@SREEHARSHA:~$ ./wait
Child running
Parent running after child
```